

$$i = 5.0 \sin 100\pi t$$

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C

$$\frac{V_s}{V_p} = \frac{5}{1} \Rightarrow V_s = 5 \times 240 = 1200 \text{ V}$$

Power delivered to the resistive load $P = \frac{V^2}{R} = \frac{1200^2}{1000} = 1440 \text{ W}$

Ideal transformer implies 100% efficiency.

Power supplied by primary circuit $P = VI = 240I = 1440 \Rightarrow I = \frac{1440}{240} = 6.0 \text{ A}$

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B